Implementation and evaluation of the reinforcement learning methods Self-Play and League Training in the MicroRTS environment

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Goals and Research Question

Goal

- improve the Base Agent with League Training (LT)
- training against a more diverse set of opponents
- focus on robustness and generalization

Base Agent

- combination of Behavioral Cloning (BC) and Proximal Policy Optimization (PPO)
- strong baseline, but weaknesses against several opponents

Research Question

- How much does LT with PPO improve robustness and generalization of the BC/PPO-pretrained agent?
- How strongly does this depend on the diversity of initial agents and BC experts?

Setting

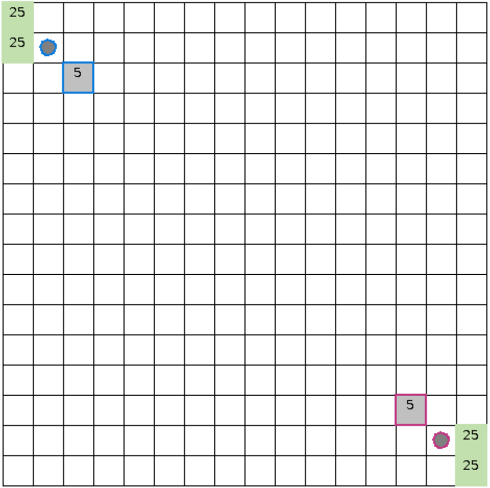
MicroRTS

- simplified RTS environment for RL research
- still contains resource management, unit production and combat
- supports controlled and repeatable experiments

Main RL challenges

- large combinatorial action space
- simultaneous decisions
- delayed and sparse rewards
- robustness against diverse opponents

▸ Open MicroRTS Example Video



Training Pipeline

BC → BC-FT → PPO → **LT** → PPO fine-tuning

BC and BC-FT

- imitate experts
- BC-FT: only winning replays

PPO

- iteratively update policy
- refine the policy through exploration

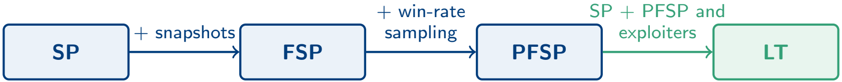
LT

- train against an evolving opponent population
- add exploiters

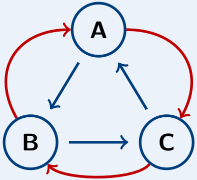
PPO fine-tuning

- repair the main remaining weakness after LT
- add WorkerRushAI as opponent

From SP to LT



cyclical learning of SP



Blue arrows: learning direction
Red arrows: policies the current policy performs well against

Why LT helps

- improves stability by identifying and exploiting weaknesses
- creates diverse set of opponents without the main agent having to find them through exploration

LT Design

Main Agent (MA)

- primary policy
- resulting agent

Historical Agents (HA)

- frozen checkpoints kept in the opponent pool
- created by all other league agent types

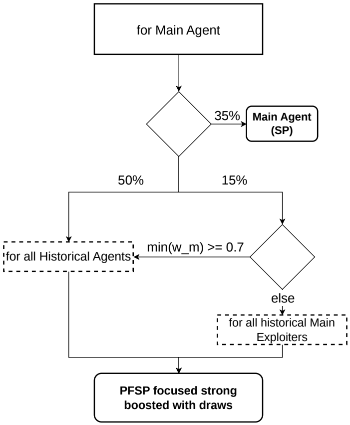
Main Exploiters (ME)

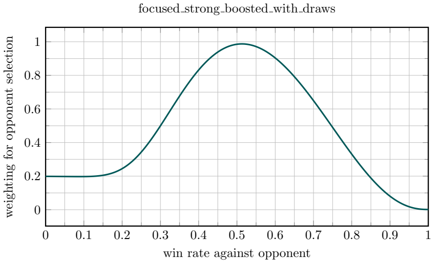
- exploit weaknesses of the MA
- reset when checkpointed

League Exploiters (LE)

- exploit broader weaknesses across the league
- can reset when checkpointed

LT MA Matchmaking



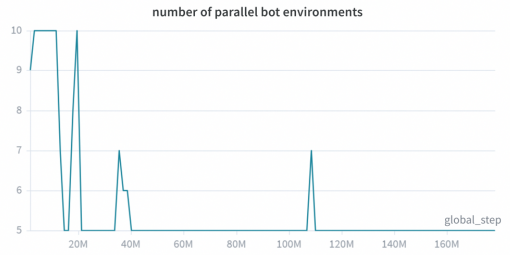


- 35% self-play
- 50% PFSP over HAs
- 15% PFSP over HMEs if win rate is low
- PFSP prioritizes unresolved matchups

Bot Environments

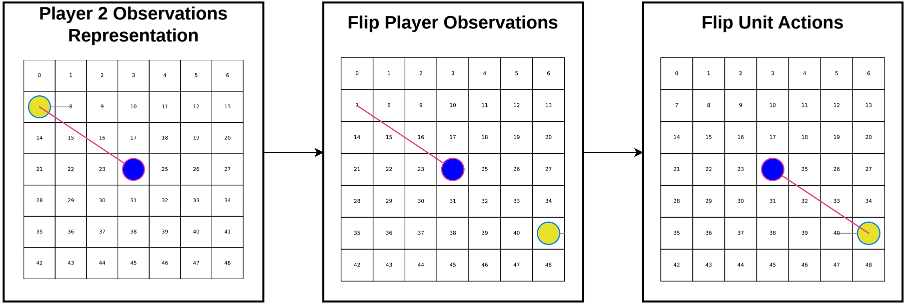
Bot environments

- additional dynamic bot environments (CoacAI and Mayari)
- scripted bots: CoacAI, Mayari, WorkerRushAI, PassiveAI, LightRushAI, RandomAI and RandomBiasedAI
- MCTS based bots: Rojo, MixedBot, Izanagi, Tiamat, Droplet, GuidedRojoA3N and NaiveMCTS



Observation adjustment

- Base Agent only trained as Player 1
- flip Player-2 GridNet observations
- flip masks and action directions



Implementation Changes

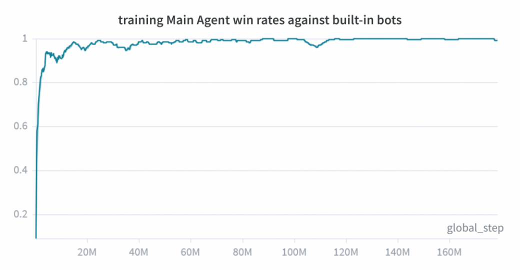
Key changes

- delta score reward: denser signal in sparse-reward games
- annealed entropy coefficient: more exploration early, less later
- new initial agents: new start points for exploiters

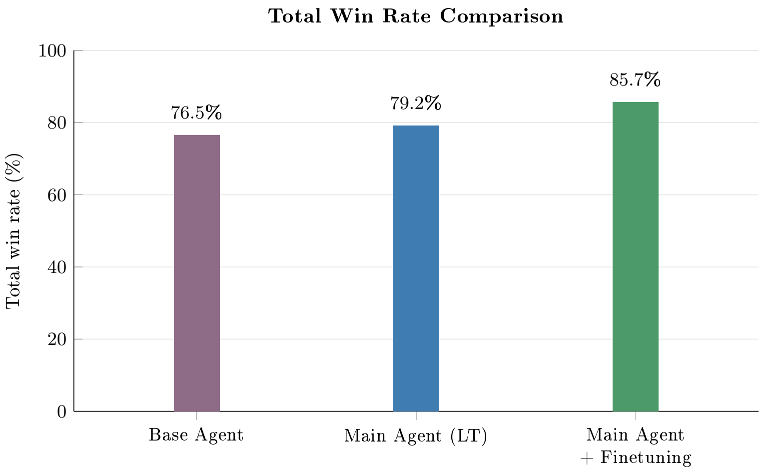
Experiment Setup

Setup

- 1 map: `basesWorkers16x16A`
- 1 MA, 4 ME, 1 LE
- for each 25 environments
- 5 to 10 dynamic bot environments
- $\sim$130M LT steps, $\sim$100 h
- max: 160 parallel environments
- eval: at least 100 games vs 12 opponents



Main Quantitative Result



Loss rate

- loss rate: **7.7%** $\rightarrow$ **2.2%**
- different Base Agent version than in the prior work

Evaluation Results (Win Rate)

Opponent	Base	LT	LT + FT
CoacAI	96.0	**100.0**	**100.0**
Mayari	67.3	**100.0**	**100.0**
WorkerRushAI	**100.0**	35.0	99.5
RandomBiasedAI	79.0	97.0	**97.5**
MixedBot	43.3	57.0	**60.5**
GuidedRojoA3N	74.7	87.0	**88.5**
Tiamat	93.0	**99.0**	**99.0**
Droplet	67.7	62.0	**75.5**
overall	76.5	79.2	85.7

How to read

- values are win rates in %
- higher is better
- bold: best value per opponent
- color: relative strength

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Training opponents

- CoacAI and Mayari in LT
- WorkerRushAI in FT

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Notable MCTS gains

- MixedBot: 43.3 $\rightarrow$ 60.5
- GuidedRojoA3N: 74.7 $\rightarrow$ 88.5
- Droplet: 67.7 $\rightarrow$ 75.5

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WorkerRushAI exploit

- LT drops to 35.0%
- FT trains against WRAI
- repaired to 99.5%

Evaluation Results (Win Rate)

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RandomBiasedAI

- LT gains remain after repair
- better against drawing behavior
- 79.0 $\rightarrow$ 97.0 after LT
- retained after FT: 97.5

Evaluation Results (Win Rate)

Opponent	Base	LT	LT + FT
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Robustness after FT

- Other examples: MixedBot, GuidedRojoA3N and Tiamat

Interpretation of the Results

Interpretation

- robustness against diverse opponents becomes part of the optimization target
- LT finds a better local optimum
- FT improves this local optimum further
- MA after LT vs Base Agent: **89%** win, **9%** draw, **2%** loss

Player 1 (P1) Priority

Effect

- P1 is prioritized in conflicts
- self-match over 200 games: P1 **38%**, draw **28%**, P2 **34%**
- visible effect, but small
- alternating the learning side prevents P1-specific behavior

▸ Open Example Video

Main Limitation: Diversity

Diversity bottleneck

- not diverse exploiters discover weaknesses with similar exploits
- strategically different exploits require unlearning Base Agents fundamental behavior first
- long transitions, bad rewards
- new initial agents $\rightarrow$ more exploiter diversity

Hypothesis

limited diversity of BC experts

↓

limited diversity of the Base Agent

↓

limited diversity for Exploiters

↓

less generalized MA

▸ Open Example Video

Remaining Limitations and Future Work

Remaining limitations

- 1 map
- full observability, no fog of war
- high memory and computation cost
- 160 LT envs vs 24 for pure PPO

Future work

- more maps
- fog of war
- MicroRTS on GPU
- stronger diversity mechanisms

Conclusion

Conclusion

- LT $+$ PPO improves robustness and generalization of the MA
- PPO fine-tuning further improves the final agent
- LT improves training and evaluation matchups
- effectiveness of LT depends on exploiter diversity
- exploiter diversity depends on initial agents and BC experts

Thank you

Questions?

Evidence

Is the improvement really caused by LT, or just by more training?

- LT alone improves the overall win rate from **76.5%** to **79.2%**.
- The final agent reaches **85.7%** after PPO fine-tuning.
- A same-compute PPO baseline would be the cleanest additional control.

What is the strongest evidence that LT changes the policy?

- MA after LT beats the Base Agent: **89%** win, **9%** draw, **2%** loss.
- Large gains against Mayari, RandomBiasedAI, Rojo and GuidedRojoA3N.

WorkerRushAI

Why does LT perform worse against WorkerRushAI?

- WorkerRushAI exposes a weakness not sufficiently found by the exploiters.
- The Base Agent is biased toward a heavy-rush strategy.
- The worker-focused initial agent for the exploiters wasn't strong enough to find it. Both of the HA created got stuch in a local optimum of drawing the Game instead.

Why does PPO fine-tuning repair it?

- WorkerRushAI is added as a direct training opponent.
- The win rate recovers from **35.0%** after LT to **99.5%**.
- Other results mostly remain stable or improve, so no strong overfitting is visible in the evaluated set.

Expected Questions: Generalization

Can we call this generalization with only one map?

- Only within the evaluated setting: one map, full observability and fixed rules.
- Generalization is tested across opponent types, not across maps.
- Several improved opponents were not used in LT training, especially MCTS-based bots.

What would be needed for a stronger claim?

- evaluation on multiple maps
- fog of war
- multiple training seeds
- same-compute PPO and ablation baselines

Expected Questions: PPO and AlphaStar

Why use PPO instead of AlphaStar's V-trace setup?

- PPO was already available in the MicroRTS/RAISocketAI training pipeline.
- It made LT practical to implement within the scope of this thesis.
- V-trace would likely scale better in a large distributed setup, but is more complex.

What is different from AlphaStar?

- smaller environment and much smaller league
- PPO instead of off-policy actor-critic with V-trace
- initial policy from BC/PPO bots, not from broad human replay data

Expected Questions: Diversity

Why is exploiter diversity the main limitation?

- Exploiters initialized from the same Base Agent tend to find similar weaknesses.
- The Base Agent strongly favors heavy-rush behavior.
- Different strategy clusters, such as worker or ranged strategies, are hard to discover from that start point.

Why add new initial agents?

- They give exploiters different strategic starting points.
- This improved difficult matchups, e. g. Droplet and GuidedRojoA3N.
- It supports the hypothesis that BC expert diversity strongly affects LT performance.

Expected Questions: Bot Environments

Why keep CoacAI and Mayari during LT?

- They stabilize the early LT phase before exploiters find useful weaknesses.
- They prevent the Main Agent from drifting away from already strong behavior.
- The number of bot environments is dynamic, so they become less dominant later.

Does this make the evaluation unfair?

- CoacAI and Mayari are training opponents, so they are not the main generalization evidence.
- The stronger evidence comes from opponents not used for LT training.

Expected Questions: Player Side

Why adjust observations for Player 2?

- The Base Agent was trained only as Player 1.
- Player-2 observations, action masks and action directions are flipped.
- This lets the same policy play both sides without relearning the map orientation from scratch.

Does Player 1 priority bias the results?

- It is a potential bias in conflict resolution.
- Self-match evaluation: P1 **38%**, draw **28%**, P2 **34%**.
- Alternating the learning side prevents purely P1-specific behavior.

Expected Questions: Draws and NaiveMCTS

Why are win rates against NaiveMCTS so low?

- NaiveMCTS often avoids losing instead of actively trying to win.
- This creates many draws and few losses.
- Therefore the loss rate is important for interpreting this matchup.

Why use loss rate in addition to win rate?

- Draw-heavy opponents can hide defensive improvements in the win rate.
- The final loss rate drops from **7.7%** to **2.2%**.
- This suggests more robust defense, not only stronger attacks.

Expected Questions: Statistical Reliability

Are the improvements statistically reliable?

- Large gains are unlikely to be explained only by evaluation noise.
- Small differences, e. g. overall LT vs Base Agent, should be interpreted cautiously.
- Only one LT training seed was used because training was expensive.

What is the cleanest future experiment?

- same compute budget for PPO, LT and LT + FT
- several random seeds
- ablations for new initial agents, bot environments, delta score and entropy annealing

Detailed Evaluation Results: Win Rates

Opponent	Main Agent without new initial agents	Base Agent	Main Agent (LT)	Main Agent + Fine-tuning
bots used in training				
CoacAI	**100.0**	96.0	**100.0**	**100.0**
Mayari	**100.0**	67.3	**100.0**	**100.0**
scripted bots				
WorkerRushAI	31.0	**100.0**	35.0	99.5
PassiveAI	**100.0**	99.7	**100.0**	**100.0**
LightRushAI	**100.0**	98.0	**100.0**	**100.0**
RandomAI	**100.0**	99.0	**100.0**	**100.0**
RandomBiasedAI	91.0	79.0	97.0	**97.5**
MCTS based bots				
Rojo	**100.0**	81.0	**100.0**	**100.0**
MixedBot	45.0	43.3	57.0	**60.5**
Izanagi	68.0	71.7	70.0	**72.5**
Tiamat	94.0	93.0	**99.0**	**99.0**
Droplet	45.0	67.7	62.0	**75.5**
GuidedRojoA3N	64.0	74.7	87.0	**88.5**
NaiveMCTS	0.0	0.7	2.0	**6.5**
overall	74.1	76.5	79.2	**85.7**

Detailed Evaluation Results: Loss Rates

Opponent	Main Agent without new initial agents	Base Agent	Main Agent (LT)	Main Agent + Fine-tuning
bots used in training				
CoacAI	**0.0**	2.0	**0.0**	**0.0**
Mayari	**0.0**	27.3	**0.0**	**0.0**
scripted bots				
WorkerRushAI	63.0	**0.0**	55.0	0.5
PassiveAI	**0.0**	**0.0**	**0.0**	**0.0**
LightRushAI	**0.0**	2.0	**0.0**	**0.0**
RandomAI	**0.0**	**0.0**	**0.0**	**0.0**
RandomBiasedAI	**0.0**	**0.0**	**0.0**	**0.0**
MCTS based bots				
Rojo	**0.0**	5.3	**0.0**	**0.0**
MixedBot	33.0	34.3	20.0	**13.5**
Izanagi	15.0	16.3	7.0	**6.0**
Tiamat	6.0	7.0	1.0	**0.5**
Droplet	22.0	**9.0**	22.0	10.5
GuidedRojoA3N	**0.0**	1.7	**0.0**	**0.0**
NaiveMCTS	**0.0**	2.7	**0.0**	**0.0**
overall	9.9	7.7	7.5	**2.2**

Main Agent vs League Exploiter

- the League Exploiter resets
- different hyperparameters
- historical League Exploiters are not included as Main Exploiter opponents

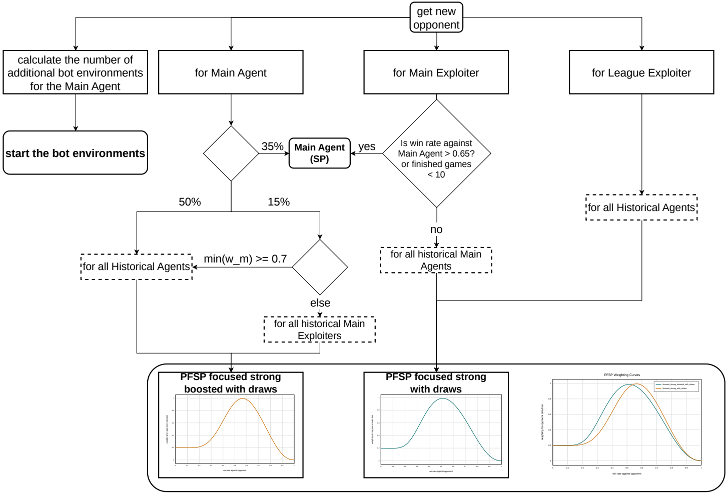
Why adjust the observations for SP?

- Gym-MicroRTS: Toward affordable full game real-time strategy games research with deep reinforcement learning implemented SP and concluded that a likely reason for worse performance of SP was that SP also has to train, to start as Player 2, while pure PPO only trained as Player 1.
- difficulties retrainig the Base Agent.

Why are the win rates against NaiveMCTS low?

- It is one of the few opponents that often gives up on winning and instead stays far away from the agent's units.
- This results in many draws and few losses.

Full LT Matchmaking



Matchmaking and opponent sampling in the LT setup.

New Initial Agents

1 agent per unit type

Heavy focused

- Base Agent

Light focused

- 9 epochs of BC with LightRushAI as the expert
- opponents for BC: Mayari, WorkerRushAI, LightRushAI

Worker focused

- 8 epochs of BC with WorkerRushAI as the expert
- opponents for BC: Mayari, WorkerRushAI, LightRushAI

Ranged focused

- 6 epochs of BC with CoacAI as the expert
- 180 updates of PPO against the Base Agent, CoacAI and Mayari
- opponents for BC: WorkerRushAI, LightRushAI, Mayari and CoacAI

Why Not Fewer Environments?

- The number of environments affects rollout size.
- Smaller rollouts make PPO updates more unstable because they provide less information about the update direction.
- More steps per rollout create similar tradeoffs to adding more environments.

win rates against Base Agent as player 2

- P1: LT-FT against P2: Base Agent: win rate 85%, draw rate 15%, loss rate 0%
- P2: LT-FT against P1: Base Agent: win rate 89%, draw rate 11%, loss rate 0%